

Rababb Pannu

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TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, CSS, SQL, Java

Frameworks: PyTorch, TensorFlow, NumPy, pandas, React, Next.js, YOLO, ROS2

Developer Tools: Supabase, Git, Docker, Node.js, Linux, AWS, S3, CI/CD

Hackathon Wins: Hack Canada, Amazon Robotics, Hack The North, UofT RSX SEEK JR, Waterloo Chess Hackathon

Certifications: AWS Certified Cloud Practitioner (Elasticsearch, EC2, ECS, Cloudwatch, Cloudfront, S3, etc)

EXPERIENCE

AI Engineering Intern

Aug. 2026 – Present

Presto Phoenix (YC S10)

San Mateo, California

- Working on the agentic voice ordering pipeline behind drive-thru deployments: streaming ASR, custom NLU and LLM orchestration, menu grounding over POS integrations, and TTS.
- Scope centers on order accuracy (NIR) and per-turn latency, including failure triage across pipeline stages and eval coverage for releases.

Software Engineering Intern

Jan. 2026 – April 2026

BMO - NARP Platforms Tech Team

Toronto, ON

- Led and developed a File Monitor System/Dashboard using DQL in Dynatrace and Python + JSON in Ansible to track file presence/metadata, SLA compliance, and send SLA risk notifications, cutting manual escalation time by **60%** and lessening SLA breaches, preventing fines.
- Helped develop a TSYS Monitoring Dashboard for end-to-end visibility into TSYS data flows, AWS API/MQ volumes, and consumer usage patterns, utilising AWS Lambda CloudWatch logs for data analysis.

Machine Learning Engineer

Sept. 2025 – Present

WATonomous

Waterloo, ON

- Trained reinforcement-learning (locomotion) and imitation-learning (manipulation) policies in Isaac Lab and PyTorch, logging rewards, trajectory error, and failure cases.
- Fine tuned a YOLOv11 machine learning computer vision model for competition specific objects, achieving **95%** accuracy and integrating it into our robotics simulation pipeline.
- Leveraged WATcloud's Slurm-managed GPU cluster to implement distributed training, reducing training time.

PROJECTS

AI Voice Persona Agent 🗣️🔗 | *PyTorch, LangGraph, Whisper, RAG, ChromaDB, HuggingFace*

- Fine-tuned OpenAI's Whisper ASR model (NLP) with LoRA(PEFT, r=32 on attention projections) on a Hugging Face technical speech dataset, then exported to CTranslate2 for real-time int8 inference, beating base Whisper
- Built a LangGraph agentic pipeline with RAG over personal data, embedding chunks with bge-small into a **ChromaDB** vector store to ground every claim in retrieved context.
- Engineered guardrails that block prompt injection, gated by a four-suite eval harness with a CI regression check.

Reparo 🛠️ | *Python, Agentic AI, Shopify Storefront API*

Hack Canada 2026

- Won \$5,000 Reactiv prize and Most Complex AI Hack finalist for an agentic pipeline that diagnoses broken household items and one-click checks out exact repair parts from live Shopify inventory (36-hour build)
- Implemented a RAG based grounded retrieval approach over real product catalogs to constrain LLM tool calls to in-stock items, eliminating hallucinated recommendations and enabling safe real-money transactions
- Architected a multi-step agent (diagnosis → part identification → catalog retrieval → multiple iterations → checkout) integrating Shopify commerce APIs for an end-to-end natural-language-to-purchase flow

FIFA World Cup Graphs Site 📊 | *Python, SQL, Matplotlib, pandas, React*

- Built an interactive data visualization platform for FIFA World Cup statistics using React with dynamic filtering and real-time graph updates.
- Engineered a backend pipeline using pandas and Matplotlib to query data and generate plots, reducing manual processing time by **95%**.

EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering

GPA: 3.3